

Student Utility Program

Aim: To understand and implement the basic programming constructs of Java, including variables, data types, operators, conditional statements, loops, and methods.

Algorithm:

- 1.Start the program.
- 2.Display menu for the operations that can be performed in the program.
- 3.Read user choice.
- 4.Use switch case to perform the selected operation.
- 5.Perform the action by executing the block of code as per user choice.
- 6.Display the output.
- 7.If wanted to continue, can continue or can be stopped.

Program code:

```
package task1 ;  
  
import java.util.Scanner;  
  
public class StudentUtilityProgram {
```

```
int factorial(int n) {  
    int fact = 1;  
    for(int i = 1; i <= n; i++) {  
        fact *= i;  
    }  
    return fact;  
}
```

```
boolean prime(int n) {  
    if (n <= 1) {  
        return false;  
    }
```

```
    for (int i=2; i <= n; i++) {  
        if (n % i == 0) {  
            return false;  
        }  
    }  
    return true;  
}
```

```
int max(int a, int b) {
```

```
    return (a > b) ? a : b;  
}
```

```
double areaOfCircle(double r) {  
    return Math.PI * r * r;  
}
```

```
public static void main(String[] args) {  
    Scanner sc = new Scanner(System.in);  
    int choice;  
    StudentUtilityProgram util = new StudentUtilityProgram();  
  
    do {  
        System.out.println("\n===== Student Utility Program =====");  
        System.out.println("1. Student Information");  
        System.out.println("2. Even or Odd");  
        System.out.println("3. Largest of Three Numbers");  
        System.out.println("4. Grade based on percentage");  
        System.out.println("5. Day of a week");  
        System.out.println("6. Multiplication Tales");  
        System.out.println("7. Numbers from 1 to N");  
        System.out.println("8. Sum of first N natural numbers");  
        System.out.println("9. Fibonacci Series");
```

```
System.out.println("10. Factorial of a number");
System.out.println("11. Prime number check");
System.out.println("12. Max of Two Numbers");
System.out.println("13. Area of Circle");
System.out.println("0. Exit");
```

```
System.out.print("Enter your choice: ");
choice = sc.nextInt();
```

```
switch (choice) {
```

```
    case 1:
```

```
        sc.nextLine(); // Consume the newline character
```

```
        System.out.print("Enter student name: ");
```

```
        String name = sc.nextLine();
```

```
        System.out.print("Enter student roll number: ");
```

```
        int rollNumber = sc.nextInt();
```

```
        System.out.print("Enter student marks in 3 subjects: ");
```

```
        int m1 = sc.nextInt();
```

```
        int m2 = sc.nextInt();
```

```
int m3 = sc.nextInt();
```

```
int totalMarks = m1 + m2 + m3;
```

```
double percentage = (totalMarks / 3.0);
```

```
System.out.println("\nStudent Information:");
```

```
System.out.println("Name: " + name);
```

```
System.out.println("Roll Number: " + rollNumber);
```

```
System.out.println("Total Marks: " + totalMarks);
```

```
System.out.println("Percentage: " + percentage);
```

```
break;
```

```
case 2:
```

```
    System.out.print("Enter a number: ");
```

```
    int num = sc.nextInt();
```

```
    if (num % 2 == 0) {
```

```
        System.out.println(num + " is even.");
```

```
    } else {
```

```
        System.out.println(num + " is odd.");
```

```
    }
```

```
    break;
```

```
case 3:
```

```
System.out.print("Enter three numbers: ");

int a = sc.nextInt();

int b = sc.nextInt();

int c = sc.nextInt();


if(a>b && a>c) {

    System.out.println(a + " is the largest number.");

} else if(b>c) {

    System.out.println(b + " is the largest number.");

} else {

    System.out.println(c + " is the largest number.");

}

break;
```

case 4:

```
System.out.print("Enter percentage: ");

double per = sc.nextDouble();


if(per >= 90) {

    System.out.println("Grade: A");

} else if(per >= 80) {

    System.out.println("Grade: B");

} else if(per >= 70) {
```

```
        System.out.println("Grade: C");
    } else if(per >= 60) {
        System.out.println("Grade: D");
    } else {
        System.out.println("Grade: F");
    }
    break;
```

case 5:

```
System.out.print("Enter day number (1-7): ");
int day = sc.nextInt();
```

```
switch(day) {
```

case 1:

```
    System.out.println("Monday");
    break;
```

case 2:

```
    System.out.println("Tuesday");
    break;
```

case 3:

```
    System.out.println("Wednesday");
    break;
```

case 4:

```
        System.out.println("Thursday");
        break;
    case 5:
        System.out.println("Friday");
        break;
    case 6:
        System.out.println("Saturday");
        break;
    case 7:
        System.out.println("Sunday");
        break;
    default:
        System.out.println("Invalid day number.");
}
break;
```

```
case 6:
    System.out.print("Enter a number for multiplication table:
");

    int n = sc.nextInt();

    System.out.println("Multiplication Table of " + n + ":");
    for(int i=1; i<=10; i++) {
        System.out.println(n + " x " + i + " = " + (n*i));
    }
}
```

```
}
```

```
break;
```

case 7:

```
System.out.print("Enter a number N: ");
```

```
int N = sc.nextInt();
```

```
System.out.println("Numbers from 1 to " + N + ":");
```

```
for(int i=1; i<=N; i++) {
```

```
    System.out.print(i + " ");
```

```
}
```

```
System.out.println();
```

```
break;
```

case 8:

```
System.out.print("Enter a number N: ");
```

```
int N1 = sc.nextInt();
```

```
int sum = 0;
```

```
for(int i=1; i<=N1; i++) {
```

```
    sum += i;
```

```
}
```

```
System.out.println("Sum of first " + N1 + " natural numbers:  
" + sum);
```

```
break;
```

case 9:

```
System.out.print("Enter a number N: ");  
  
int N2 = sc.nextInt();  
  
int x = 0, y = 1;  
  
System.out.println("Fibonacci Series up to " + N2 + ":");  
while(x <= N2 ) {  
    System.out.print(x + " ");  
    int next = x + y;  
    x = y;  
    y = next;  
}  
  
System.out.println();  
  
break;
```

case 10:

```
System.out.print("Enter a number: ");  
  
int N3 = sc.nextInt();  
  
int fact = util.factorial(N3);  
  
System.out.println("Factorial of " + N3 + " is: " + fact);  
  
break;
```

case 11:

```
System.out.print("Enter a number: ");  
  
int N4 = sc.nextInt();  
  
boolean isPrime = util.prime(N4);  
  
if(isPrime) {  
    System.out.println(N4 + " is a prime number.");  
} else {  
    System.out.println(N4 + " is not a prime number.");  
}
```

case 12:

```
System.out.print("Enter two numbers: ");  
  
int N5 = sc.nextInt();  
  
int N6 = sc.nextInt();  
  
int max = util.max(N5, N6);  
  
System.out.println("Maximum of " + N5 + " and " + N6 +  
" is: " + max);  
  
break;
```

case 13:

```
System.out.print("Enter radius of the circle: ");  
  
double r = sc.nextDouble();  
  
double area = util.areaOfCircle(r);
```

```

        System.out.println("Area of the circle with radius " + r +
" is: " + area);

        break;

    case 0:

        System.out.println("Exiting the program.");

        break;

    default:

        System.out.println("Invalid choice. Please try again.");

    }

} while (choice != 0);

sc.close();

}

}

```

Output:

Sample Output

===== Student Utility Program =====

1. Student Information
2. Even or Odd
3. Largest of Three Numbers
4. Grade based on percentage

5. Day of a week
6. Multiplication Tales
7. Numbers from 1 to N
8. Sum of first N natural numbers
9. Fibonacci Series
10. Factorial of a number
11. Prime number check
12. Max of Two Numbers
13. Area of Circle
0. Exit

Enter your choice: 1

Enter student name: Rahul

Enter student roll number: 101

Enter student marks in 3 subjects: 80 75 90

Student Information:

Name: Rahul

Roll Number: 101

Total Marks: 245

Percentage: 81.66666666666667

Enter your choice: 2

Enter a number: 17

17 is odd.

Enter your choice: 3

Enter three numbers: 25 80 45

80 is the largest number.

Enter your choice: 4

Enter percentage: 87

Grade: B

Enter your choice: 5

Enter day number (1-7): 3

Wednesday

Enter your choice: 6

Enter a number for multiplication table: 5

Multiplication Table of 5:

$$5 \times 1 = 5$$

$$5 \times 2 = 10$$

$$5 \times 3 = 15$$

$$5 \times 4 = 20$$

$$5 \times 5 = 25$$

$$5 \times 6 = 30$$

$$5 \times 7 = 35$$

$$5 \times 8 = 40$$

$$5 \times 9 = 45$$

$$5 \times 10 = 50$$

Enter your choice: 7

Enter a number N: 5

Numbers from 1 to 5:

1 2 3 4 5

Enter your choice: 8

Enter a number N: 5

Sum of first 5 natural numbers: 15

Enter your choice: 9

Enter a number N: 10

Fibonacci Series up to 10:

0 1 1 2 3 5 8

Enter your choice: 10

Enter a number: 5

Factorial of 5 is: 120

Enter your choice: 11

Enter a number: 13

13 is a prime number.

Enter your choice: 12

Enter two numbers: 45 70

Maximum of 45 and 70 is: 70

Enter your choice: 13

Enter radius of the circle: 5

Area of the circle with radius 5.0 is: 78.53981633974483

Enter your choice: 0

Exiting the program.

Theory Questions and Answers

Q1. What is the structure of a Java program?

A Java program normally consists of the following parts:

1. Package declaration – optional; groups related classes.
2. Import statements – optional; bring classes such as Scanner into the program.
3. Class declaration – defines the class.
4. Data members and methods – contain variables and operations.
5. main() method – the starting point for a standalone Java application.
6. Statements – instructions executed by the program.

Example structure: package → imports → class → methods → main() → statements.

Q2. Differentiate between primitive and non-primitive data types.

Primitive data types are the basic built-in types in Java. Examples:

byte, short, int, long, float, double, char and boolean.

They store simple values directly.

Non-primitive/reference data types include classes, objects, arrays, interfaces and String. A reference variable refers to an object or data stored elsewhere.

In this program, int, double and boolean are primitive types, while String is a non-primitive/reference type.

Q3. Compare if-else and switch statements.

if-else: Best for conditions involving ranges, relational operators and complex logical expressions. It evaluates boolean conditions.

switch: Best when one expression is compared against several fixed

values/cases. It makes menu-based programs easier to read.

Example in this program: case 5 uses switch for day numbers, while case 4 uses if-else because grades depend on ranges such as 90, 80, 70, etc.

Q4. Explain the differences between for, while, and do-while loops.

for loop: Generally used when the number of iterations is known.

Example: printing 1 to N and multiplication tables.

while loop: Used when repetition depends on a condition and the number of iterations may not be known beforehand.

Example: Fibonacci series in case 9.

do-while loop: Executes the body at least once because the condition is checked after the body. This program uses it to display the menu at least once and continue until choice 0.

Basic forms: for(initialization; condition; update), while(condition), and do { } while(condition);

Q5. What are methods? Why are they useful in Java?

A method is a named block of code that performs a particular task. A method may accept parameters and may return a value.

Methods are useful because they provide code reusability, modularity, readability, easier testing and easier maintenance.

In this program, factorial(), prime(), max() and areaOfCircle() are methods. For example, util.factorial(N3) calls the factorial method instead of writing the factorial logic again inside main().

Q6. Explain the difference between formal parameters and actual parameters.

Formal parameters are the variables written in a method definition. They receive values when the method is called.

Example: `int max(int a, int b)` – a and b are formal parameters.

Actual parameters (arguments) are the values or variables supplied during the method call.

Example: `util.max(N5, N6)` – N5 and N6 are actual parameters.

Therefore: Method definition → formal parameters; Method call → actual parameters.